ZIHENG ZHANG

3044 Veteran Ave, Los Angeles, California, 90034 | 310-625-1831 | zihengz1211@g.ucla.edu

EDUCATION

University of California, Los Angeles — **Los Angeles, USA**

Master of Science in Biostatistics — *September 2023 – Present*

- Relevant Coursework: Methods in Biostatistics C (A+), Mathematical Statistics B (A+), Mathematical Methods for Biostatistics (A+)
- GPA: 4.00/4.00

Xi'an Jiaotong-Liverpool University — **Suzhou, China**

Bachelor of Science in Applied Mathematics — *September 2019 – July 2023*

- Relevant Coursework: Introduction to Probability and Statistics Applied Probability (4.00/4.00), Statistical Distribution Theory (4.00/4.00), Applied Probability (4.00/4.00)
- GPA: 3.96/4.00 (WES); GRE: 334; IELTS: 7.5; TOEFL: 94

PUBLICATIONS/PREPRINTS

- Wang, C. H., **Zhang, Z. H.**, and Li, J. J. (2025). SyNPar: Synthetic Null Data Parallelism for High-Power False Discovery Rate Control in High-Dimensional Variable Selection. arXiv:2501.05012 [stat.ME]. Available: https://arxiv.org/abs/2501.05012

RESEARCH EXPERIENCES

Explore Statistical Method for FDR Control in Genomic Data — **Los Angeles, USA**

Graduate Student Researcher — *September 2024 – Present*

Advisor: Prof. Jingyi Jessica Li, Dept of Biostatistics, UCLA

- **Methodology Integration and Code Reproduction:** Integrate our method with existing knockoff and data-splitting methods for comparison; reproduce, adapt, and fine-tune the code from relevant articles to suit our dataset and objectives.
- **Simulation and Testing:** Run simulations to evaluate the method's performance across various scenarios, varying key parameters to optimize robustness and accuracy.
- **Visualization and Documentation:** Generate plots to visualize results and thoroughly write the simulation section of the paper, documenting the methodology, parameter variations, and comparative outcomes.

Exploring Brain Functional Connectivity of FTD Patients in fMRI Data — **Los Angeles, USA**

Graduate Student Researcher — *April 2024 – Present*

Advisor: Prof. Michele Guindani, Dept of Biostatistics, UCLA

- **Data Preprocessing:** Employ SPM12 and GIFT (Group ICA Toolbox) to preprocess resting-state fMRI data, preparing it for further analysis by enhancing data quality and reliability.
- **Dynamic Functional Network Connectivity Analysis**: Analyze dynamic functional network connectivity to assess differences in functional connectivity between FTD subgroups and healthy controls.
- **Bayesian Approach Development**: Develop a Bayesian approach, incorporating Gaussian graphical models and causal mediation analysis, to explore the impact of GRN mutations on brain connectivity and related cognitive and behavioral outcomes.

Optimization of an Experimental Design for Dose-Response Modeling — **Los Angeles, USA**

Graduate Student Researcher — *June 2024 – Present*

Advisor: Prof. Weng Kee Wong, Dept of Biostatistics, UCLA

- **Code Reproduction and Accuracy Verification**: Reproduce the code from a two-stage experimental design for dose-response modeling and compare the results with the original article to ensure accuracy and reliability.

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- **Simulation Modification**: Adjust simulation settings and parameters from the original article to assess result variability; evaluate the robustness and sensitivity of the two-stage design under different conditions.
- **Design Improvement and Optimization**: Develop enhancements to the two-stage experimental design based on insights from code reproduction and robustness testing to improve performance and efficiency.

Research on Fairness in Machine Learning — **Suzhou, China**
Undergraduate Student Researcher — *June 2023 – August 2023*
Advisor: Prof. Yi Yang, Dept of Economics, XJTLU

- **Article Integration on Classification Methods**: Collected and synthesized key articles on binary and multiclass classification to establish a foundation for analysis.
- **Classifier Fairness Comparison**: Compare the fairness of the project's classifier with baseline methods, reweighing, disparate impact remover, and equalized odds, by using standard fairness metrics.

Developing New Time-to-Event Phase I Clinical Trial Design — **Suzhou, China**
Undergraduate Student Researcher — *July 2022 – May 2023*
Advisor: Prof. Xiaojun Zhu, Dept of Financial and Actuarial Mathematics, XJTLU

- **Review of Phase I Clinical Trial Designs**: Conduct a thorough analysis of Phase I trial designs, summarizing findings in a poster presentation during the undergraduate research fund program at XJTLU.
- **CRM and Related Design Review**: Review the continual reassessment method (CRM) and adaptive trial designs, gaining expertise in Phase I clinical trials.
- **Trial Design for Late-Onset Toxicity**: Develop a novel phase I design using Bayes' theorem to address late-onset toxicity, reducing trial duration by 45%, improving dose accuracy by 10% and safety by 15%.

Integrated study of Biostatistics and Data Analysis — **New York, USA**
Undergraduate Student Researcher — *February 2022 – May 2022*
Advisor: Prof. R. Todd Ogden, Dept of Biostatistics, Columbia University

- **Application of Statistical Methodologies**: Apply advanced statistical tools in complex research, performing in-depth exploratory analyses, linear regression, and time series analyses.
- **Public Health Applications**: Lead a group study on obesity and anthropometric data, applying statistical methods to public health issues.

WORK EXPERIENCE

Suzhou Center for Disease Control and Prevention (CDC) Biobank — **Suzhou, China**
Clinical Trial Assistant — *August 2022 – September 2022*

- Recorded the data of a clinical trial on insulin efficacy and pH value
- Analyzed the relationship between the efficacy and the pH values

HONORS

- 2020 XJTLU "Merit Students"
- 2019/2020 University Academic Achievement Award
- 2020/2021 University Academic Achievement Award
- 2021/2022 University Academic Achievement Award

SKILLS

- Proficient in MATLAB, R, Python, SAS
- Familiar with machine learning algorithms and deep learning frameworks in PyTorch
- Experience with data analysis and visualization tools (e.g., Pandas, Numpy, Matplotlib)